



ZEN



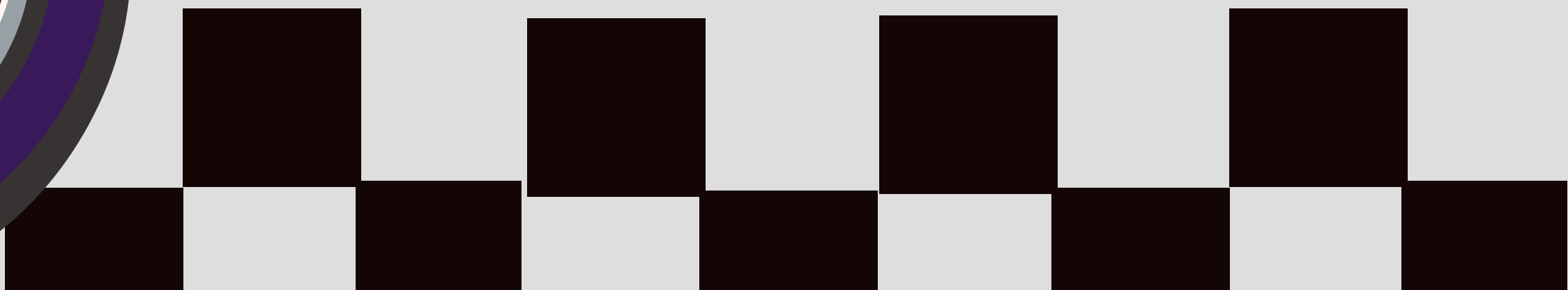
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TEAM INTRODUCTION



M. RIZKY A



CARLsoon K



QAIS ISMAIL



SABBIA MPD



LATAR BELAKANG & PERMASALAHAN

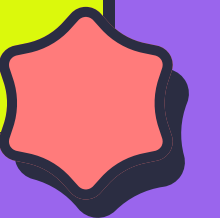
Pencatatan rental masih manual (kertas/spreadsheet)
awan human error, data tidak sinkron

Tidak ada pencatatan maintenance terpusat
mobil rusak bisa tetap disewakan

Tidak ada tracking lokasi kendaraan
keamanan aset rendah, sulit pantau mobil yang sedang disewa

Pembayaran tidak tercatat digital
sulit rekap keuangan, tidak ada audit trail

Perhitungan denda manual
potensi dispute, inkonsistensi tarif



ZENAUTO



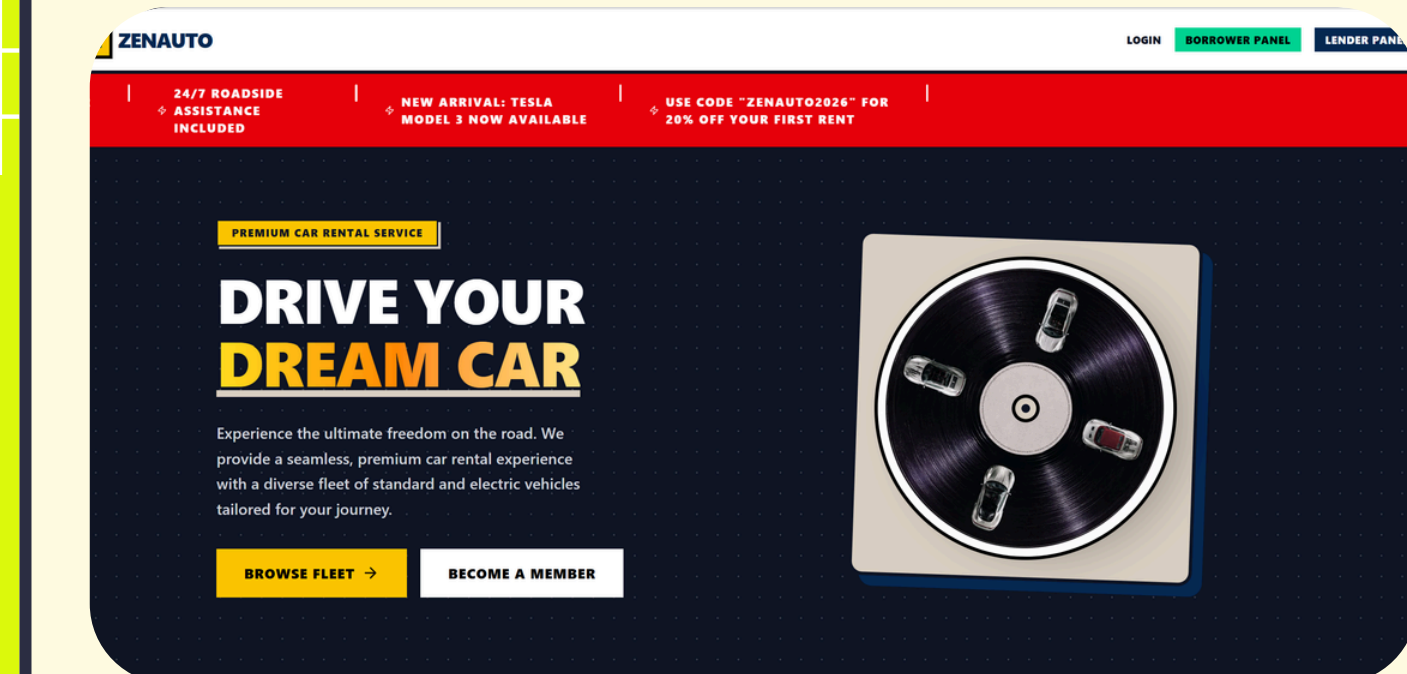
Aplikasi web manajemen **rental kendaraan** dengan dua panel utama:

Borrower (Penyewa):

- Cari & filter mobil available
- Reservasi + kunci harga
- Pembayaran multi-method
- Gunakan mobil (GPS tracked)
- Kembalikan -> auto cek denda

Lender (Pemilik):

- Kelola armada (CRUD)
- Catat maintenance/servis
- Pantau denda dari borrower
- Rekap pendapatan
- GPS Live Tracking di peta



Register/Login



Pilih Role



Borrower Flow / Lender Flow



Dashboard & History



FITUR

NO	FITUR	DESRRIPSI	TECH
1	Reservasi & Price Lock	Harga dikunci saat booking —tidak berubah walau harga model diupdate.	PG Transaction
2	Pembayaran	Mulai method, auto update. status transaksi & mobil yang rented	PG Transaction
3	Denda Otomatis	Bandingkan actual return vs. end date → jika telat auto generate penalty	PG Conditional
4	Maintenance	Mobil dikirim ke bengkel → status maintenance.	PG Transaction
5	GPS Live Tracking	Pencatatan lokasi berkala. ditampilkan di peta interaktif	MangoDB Bucket
6	Upload Gambar	Foto mobil diupload via form disimpan di cloud storage	Supabase Storage
7	FAQ Interaktif	Customer inquiry — Adminiswab → auto-publish di Landing Page	PG Static Flow
3	Add-ons	Opsi supir & asuransi tambahan saat booking	ISDDB Column

TECH STACK

Frontend

- **Vite + React** (TypeScript/TSX) : build tool + UI
- **Tailwind CSS** + PostCSS: Neo-Brutalism theme
- **React Router DOM**: SPA routing
- **React Hooks + Context** : state management & auth
- **React Leaflet** + OpenStreetMap: peta GPS

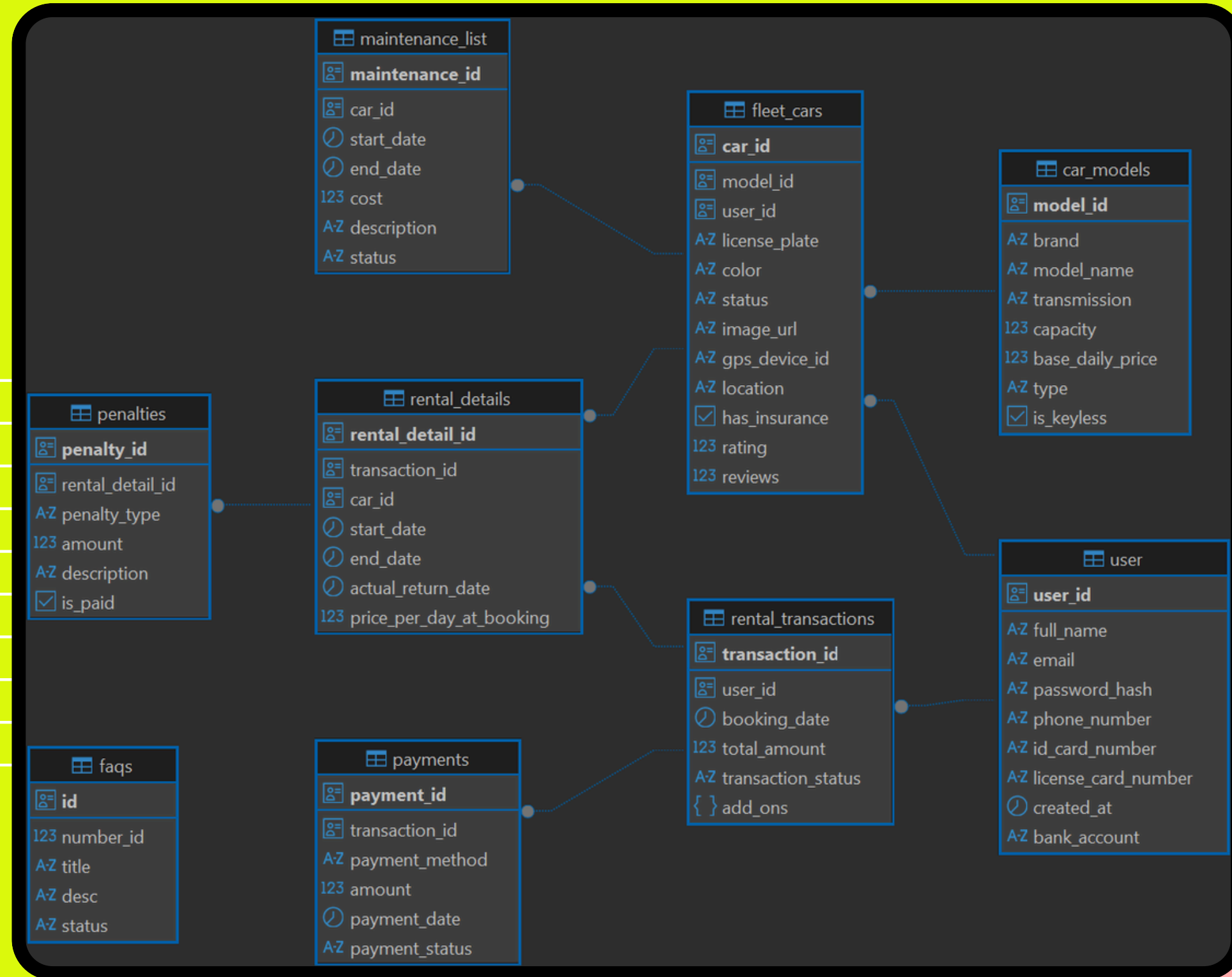
Backend

- **Node.js + Express.js** : runtime & framework
- **bcrypt** : hashing password
- **JWT** : autentikasi stateless via cookie
- **Multer**: upload file ke memory
- **Mongoose**: ODM untuk MongoDB

DATABASE / CLOUD	TEKNOLOGI	ALASAN PEMILIHAN
 Database Utama	PostgreSQL (Supabase)	✓ ACID-compliant, cocok untuk data transaksional rental
 Database Pendukung	MongoDB Atlas	✓ Document-based, cocok untuk time-series GPS high-frequency
 Image Storage	Supabase Storage	✓ Integrasi langsung, public URL otomatis
 Hosting	Vercel	✓ Gratis, support serverless + static hosting

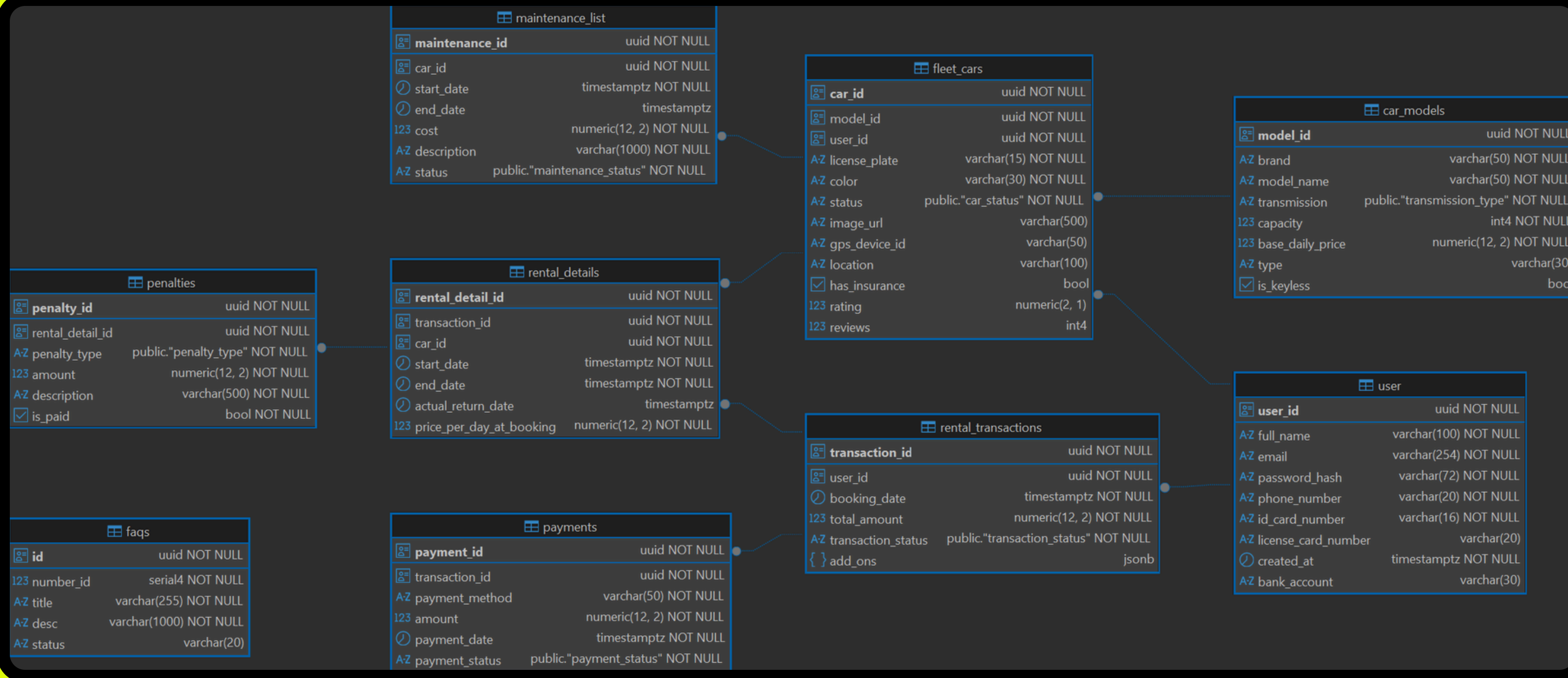


UML





ERD



FLOWCHART



API ENDPOINTS AUTH & BORROWER

Auth & User

NO	METHOD	ENDPOINT	FUNGSI
1	POST	/api/auth/register	Registrasi user baru (hash password bcrypt)
2	POST	/api/auth/login	Login, generate JWT token ke cookie
3	GET	/api/users/profile	Ambil data profil dari JWT token
4	PUT	/api/users/profile	Update SIM & rekening bank

Borrower

NO	METHOD	ENDPOINT	FUNGSI	TX?
5	GET	/api/borrower/cars/available	Katalog mobil (JOIN fleet_cars + car_models)	—
6	POST	/api/borrower/reservations	Buat reservasi, kunci harga	TX
7	GET	/api/borrower/reservations	Riwayat transaksi borrower	—
8	POST	/api/borrower/payments	Pembayaran, update 3 tabel sekaligus	TX
9	POST	/api/borrower/returns/:id	Pengembalian + auto-penalty jika telat	TX

API ENDPOINTS LENDER, TELEMETRY, FAQ

Lender

NO	METHOD	ENDPOINT	FUNGSI	TX?
10	GET	/api/lender/car-models	Master model mobil (dropdown)	—
11	GET	/api/lender/fleets	List armada milik lender	—
12	POST	/api/lender/fleets	Tambah unit mobil baru	—
13	PUT	/api/lender/fleets/:id/withdraw	Tarik mobil (cek status dulu)	—
14	POST	/api/lender/maintenances	Kirim mobil ke maintenance	TX
15	GET	/api/lender/maintenance	Riwayat maintenance	—
16	PUT	/api/lender/maintenances/:id/complete	Selesaikan maintenance	TX
17	GET	/api/lender/penalties	Daftar denda dari borrower	—
18	GET	/api/lender/finances	Rekap pendapatan + denda	—
19	GET	/api/lender/dashboard	Statistik agregat dashboard	—

Telemetry & Upload

NO	METHOD	ENDPOINT	FUNGSI	DB
20	POST	/api/telemetry/locations	Simpan koordinat GPS ke bucket	MongoDB
21	GET	/api/telemetry/locations/:car_id	Ambil lokasi terbaru	MongoDB
22	POST	/api/uploads/car-image	Upload foto → Supabase Storage URL	Supabase

FAQ

NO	METHOD	ENDPOINT	FUNGSI
23	GET	/api/faqs	Ambil FAQ yang sudah approved
24	POST	/api/faqs/inquiry	Customer kirim pertanyaan (status: pending)
25	PUT	/api/faqs/publish/:id	Admin jawab & publish → approved

HALAMAN FRONTEND

Halaman Publik

HALAMAN	ROUTE	DESKRIPSI
Landing Page	/	Promosi, fitur, CTA, FAQ interaktif, Chat Bubble
Register	/register	Form registrasi akun baru
Login	/login	Form login → JWT token
Profile	/profile	Lengkapi SIM & rekening bank

Lender Panel

HALAMAN	ROUTE	DESKRIPSI
Dashboard	/lender	Statistik armada, pendapatan, status
Fleet	/lender/fleets	CRUD armada + tambah/tarik unit
Maintenance	/lender/maintenance	Catat servis + biaya + selesaikan
Finances	/lender/finances	Pendapatan + daftar denda
Live Tracking	/lender/tracking/:car_id	Peta Leaflet GPS tracking

Borrower Panel

HALAMAN	ROUTE	DESKRIPSI
Dashboard	/borrower	Ringkasan sewa aktif & tagihan
Browse Cars	/borrower/cars	Katalog + gambar + filter
Car Detail	/borrower/cars/:car_id	Detail + date picker + kalkulator + add-ons
Checkout	/borrower/checkout/:tx_id	Konfirmasi pembayaran
History	/borrower/history	Riwayat transaksi + tombol Return

Utility

HALAMAN	ROUTE	DESKRIPSI
GPS Simulator	/simulator/gps	Hidden route buka di HP, kirim lat/ing ke MongoDB

JOBDESK



Carlsson - Backend 1

- Setup koneksi PostgreSQL (Supabase) & MongoDB Atlas
- Schema SQL (DDL) & database dump
- Mongoose model untuk GPS telemetry
- Middleware: JWT auth, Multer upload, Error handler
- Auth controller & routes (register, login, profile)
- Entry point Express (index.js)



Alam - Backend 2

- Borrower controller & routes (5 endpoint)
- Lender controller & routes (10 endpoint)
- Telemetry controller & routes (2 endpoint)
- Upload controller & routes (1 endpoint)
- FAQ controller & routes (3 endpoint)
- Deploy backend ke Vercel



Sabbia - Frontend 1

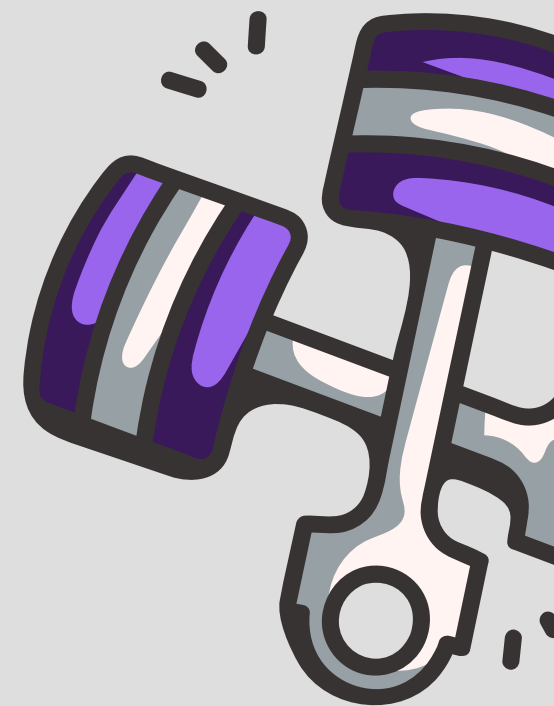
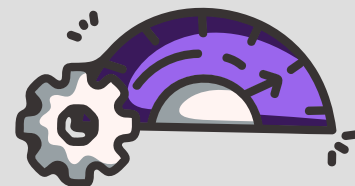
- Setup Vite + React + Tailwind + PostCSS + TypeScript
- AuthContext.tsx (global auth state)
- TypeScript type definitions
- Halaman publik: Landing, Login, Register
- Seluruh Borrower Panel
- Integrasi API Auth & Borrower
- PPT Laporan Presentasi



Qais - Frontend 2

- Membuat relational database
- Seluruh Lender Panel
- GPS Simulator page
- Integrasi React Leaflet + OpenStreetMap
- App.tsx routing
- Deploy frontend ke Vercel





THANK YOU

